

ATNLP Exam Review Guide 2026

(Some parts of this exam guide were adapted from older NLU and ANLP exam guides)

Shay Cohen, Lexi Birch and Alessandro Suglia

1 Exam content and format

- The final exam will cover content from the whole course.
 - That includes materials from lectures (excluding guest lectures), assigned readings, tutorial and lab exercises and the coursework assignment.
 - The exam will include some questions that require basic mathematical understanding and fluency with concepts presented in the class, and some that require a higher level analysis of a situation or a circumstance in modern NLP. (For more discussion of question types, see the Revision Tips below.)
 - This course covers a lot of material. If you are struggling with that, we suggest focusing on the core material that is covered in detail in the lectures, coursework, and tutorials.
- Questions will be worth 50 marks in total and all questions will be marked (you don't get to choose which ones to answer). The marks are divided into three questions. The three questions might include several thematic subquestions and in addition, a few multi-choice questions (not necessarily related to the main theme).
- Calculators and dictionaries are not permitted.
- You will write your answers on paper in a provided exam booklet.

2 Revision tips

- **Don't wait until the last minute.** Study early and often. Research shows that spaced repetition is the most effective way to remember material.
- **Study actively.** Simply reading information is not a good way to remember or understand it. Work through examples, making sure you get the same answer that's in the text or notes, and think about what questions we might ask related to that material (see below about types of questions). Also ask yourself how different parts of the course connect to each other. What are the different themes? What are different ways to approach some of the tasks?
- **Consider the types of questions** we might ask, and how they would apply to different parts of the course. Types of questions include:
 - **Bookwork** questions, which just require you to remember facts or definitions. We rarely have many marks associated with this type of question, because simply remembering a fact doesn't demonstrate that you understand how to apply it appropriately, and in a real-life situation you would typically be able to look up facts as needed. However, such questions might be a lead up for further analysis of memorised definitions or facts.
 - **Applying** a method, model or language concept to a particular situation.

- **Synthesising** knowledge to address a new situation. For example, given a new task related to one we’ve discussed, discuss how you could apply or extend methods we have seen to address it, and the problems you might expect to encounter in doing so.
- **Multiple-choice** where you are required to think in a very narrow way on a specific concept or otherwise, and respond to a focused question about it.
- **Practice the way you will be tested.** You will need to write your answers in pen on paper, in a limited amount of time. So practice doing that, using questions from past papers, tutorial exercises, etc.
- The course is new, so there are no previous exams from it. However, practice exam questions. Below we include some reference to exams with relevant questions in style and content.
- Try to estimate how long it takes you to solve one of these questions by simulating real exam conditions. You will be able to infer from that what to expect from the actual exam.
- **Use the tools you learned about!** This class places significant emphasis on LLMs. Make use of the tools you’ve learned about to actively engage with the material. You can upload slides or other resources and analyse them using current AI tools, provided this does not interfere with your own understanding and absorption of the content. For example, you might ask an AI tool to generate practice questions, summarise sections, or clarify parts that are unclear. Note: AI systems are still effectively in a “beta” stage. As discussed in class, they represent one of the largest and most widespread forms of beta testing we have experienced. Use them with appropriate caution.

3 Past papers

Past papers are available at this link. They cover previous years starting in 2014. You should *not* expect to be able to answer questions about material that was not covered in the course. If in doubt, ask the lecturers or ask on Piazza during your revision.

Example questions from previous exams that might be relevant to this course:

- NLU 2025 (<https://edin.ac/3P2Ju4n>) - Q1, Q2, Q3 g), j)
- NLU 2024 (<https://edin.ac/47d1jE7>) - Q2b, Q3
- NLU 2023 (<https://edin.ac/3Mshk28>) - Q3

4 Exam-taking tips

- **Plan your time and use it wisely.** You’ve got 120 minutes to answer 50 marks worth of questions, so think ahead about how you want to use that time.
 - You’ll probably want to spend a few minutes at the beginning to skim the questions and plan your approach, and a few minutes at the end to review your answers or come back to questions you did not fully answer.
 - After accounting for that, you’ll probably have, on average, around 2 minutes per mark. However, some questions will be much quicker to answer than others. Since you don’t need to answer questions in any particular order, you may want to start with the ones that you find quick and easy. You could also start with a partial or basic version of your answer, and leave some space to fill in more details later if you have time. (But see the next several tips.)

- **Keep track of your progress.** For example, you can tick off each question (or question part) when you’ve done it, or make a note if you’ve written something for that question but you want to come back to it if you have time.
- **Look for easy parts.** . Where a question has multiple parts, *often* the earlier parts are easier than the later parts, but not always!
 - Some multi-part questions have a “narrative” or natural ordering that doesn’t align well with moving from easy to hard. So do read all parts of the question: even if you can’t answer one part, you still may be able to answer the next part.
 - The number of marks on a question is only weakly correlated with its difficulty, because if all the hard questions were worth a lot, most students would do very poorly! Our goal is to achieve a spread of marks that shows how well each student has met the learning outcomes. This also means that most high-value questions have ways to achieve partial credit, and most questions where partial credit is not possible (e.g. multiple-choice questions) are not worth as much.
- **Answer the question that is asked.** Read each question carefully, and make sure you understand what it’s asking, and what type of answer is needed. For example, if the question asks you to justify your answer, you will get little or no credit for simply stating the answer, even if it is correct; you will need to provide a reason.
- **Don’t write more than needed.** There are at least two potential failure modes to avoid:
 - If a single word or phrase can clearly demonstrate you know the answer, don’t waste time writing full sentences or repeating parts of the question.
 - If (say) a single reason or a brief explanation is enough to answer the question, don’t add more just for the sake of it. And in particular, if you don’t know the answer, please don’t tell us everything you know about the topic of the question just in case you stumble on the right answer. Answers that include a lot of irrelevant information indicate to us that you do not have a strong grasp of the relevant concepts, so you are not doing yourself any favors and may be using up valuable time.

In some cases, you might be unsure how much detail is needed. Most questions have hints to help you, but if you’re still not sure, consider providing an initial answer with the most important information, and coming back to it later to add more detail if you have time (and if those extra details are actually relevant).

- **Do demonstrate your understanding clearly.** As noted above, a single word or phrase can sometimes be enough—but not always! Take note of any hints provided, and make sure you’ve expressed yourself clearly. We don’t deduct marks for minor problems with spelling or grammar, but your English does need to be good enough to convince us that you know the right answer and can explain it clearly. (Note: your job is to convince us. It is not our job to guess whether you might know the right answer given what you wrote. You do not get the benefit of the doubt.)

5 ATNLP 2026 examinable topics

Below is a list of concepts you should be familiar with and questions you should be able to answer if you are thoroughly familiar with the material in the course. It is safe to assume that if you have a good grasp of everything listed here, you will do well on the exam. However, we

will not guarantee that only the topics mentioned here, and nothing else, will appear on the exam.

Please note that due to time constraints during the lectures, some parts of the material is not examinable (from the slides). This includes:

- Lecture 10 (Distillation) - Slide 27 to the end.
- Lecture 13 (Reasoning in LLMs) - Slide 25 to 29.
- Lecture 16 (Mechanistic interpretability) - Slide 29 to the end.
- Lecture 17 (LLMs and Automata) - Slide 28 to the end.
- Lecture 20 (Safety and Security) - Slide 19 to the end.
- Lecture 28 (Agents 1) - Slide 27 to Slide 34.

Please note that in cases the *required* reading material covered some of these slides, the material is still examinable, but perhaps as portrayed by the reading material rather than as portrayed by the slides.

Topics:

- Machine translation: understand challenges, what data is used and how we extract it, encoder-decoder models, human and automatic evaluation
- Question Answering: understand reading comprehension, datasets like SQuAD and natural questions, Open domain QA, Fusion in decoder
- Data and Synthetic Data: importance, main data sources, understand types of data selection, deduplication, how FineWeb put this into practice, how to create synthetic instruction data, ethics of data
- Evaluation: evaluation frameworks, intrinsic evaluation, benchmarks, saturation, examples of evaluation benchmarks, LM Arena, ELO, LLM-as-a-judge, Contamination
- Long context modelling: understand sinusoidal and rotary position embeddings, positional interpolation, YaRN, efficient attention, evaluation of long context
- RAG: retriever-reader architecture, difference between parametric and non-parametric memory, sparse and dense retrieval, efficient retrieval, RAG and its variants, GraphRAG
- Multilingual LLMs: advantages and challenges, FineWeb2, data mix, understand basic ATLAS scaling laws for multilingual models, challenges of multilingual evaluation, responsible multilingual modelling
- Bias and ethics: importance and taxonomy of risks, measuring and mitigating bias
- Understand basic issues and concepts about the safety of AI models (including watermarking and adversarial attacks).
- Understand instruction finetuning, reinforcement learning from human feedback, very basic RL (all based on the coursework).
- Understand the basic concepts, tasks, models and evaluation of summarisation.
- Understand parameter efficient finetuning and algorithms in the context of parameter composition.

- Understand the different types of methods and approaches to distillation.
- Understand what reasoning means and what are the different approaches to it, especially during decoding.
- Understand mechanistic interpretability and the central concepts that arise with it.
- Understand how syntactic parsing works, its motivation, parsing structures and parsing algorithms.
- Understand the motivation behind Perceptually-grounded NLP (visual grounding, communicative grounding, symbol grounding problem)
- Understand the different methods to build vision-language models using a backbone LLM
- Understand the importance of the project/connector in the adaptation process of a VLM
- Understand the training data used to build VLMs with a focus on SmolVLM
- Benchmarking VLMs: intrinsic vs extrinsic evaluation
- Understand what an AI agent is and what the main components of its architecture are
- Understand how to build reliably and robust agentic solutions based on text-only LLMs
- Understand how to go beyond text-only agents with vision-based agentic solutions
- Understand the training data required for vision-based agents (i.e., trajectories)
- Training and evaluation of vision-based computer use agents with a focus on OpenCUA